

Miguel Romero

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EDUCATION

Pennsylvania State University

Bachelor of Science in Computer Science

University Park, PA

Aug. 2022 – May 2026

- **Relevant Coursework:** Database Management Systems, Machine Learning and Algorithmic AI, Artificial Intelligence, Exploratory Data Mining, Intro to Cryptography

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C, C++, Java, Verilog, PLC

ML & AI: PyTorch, NumPy, Pandas, scikit-learn, Ollama, LM Studio

Backend & Data: FastAPI, SQLAlchemy, Pydantic, SQLite, Power BI, MS Project, REST APIs

Cloud & Tools: AWS, Azure AI, Docker, Git, n8n, CODESYS, FACTORY I/O, NVIDIA Isaac Sim, VS Code

PROJECTS

Dual-Cell Production Line Balancer | *CODESYS (Structured Text), FACTORY I/O*

2026

- Programmed independent state machines in CODESYS Structured Text for two machining centers (Idle → Load → Machine → Unload) against FACTORY I/O's Production Line scene.
- Built a supervisory controller that tracks running lid and base counts and throttles conveyor feed to the trailing machine so neither machining center gets far ahead of the other.
- Added sensor-timeout watchdogs on each station to auto-fault and log a jam instead of stalling the line silently, and logged every cycle (station, part type, cycle time, fault state) for a full audit trail.

Plant Operations Automation | *Python, SQLite, n8n, Power BI, Webhooks, Local LLM*

2026

- Modeled a 24/7 plastic-film facility in a 16-table SQLite schema covering 293 orders, 809 logged rolls, and 1,800 ledger rows across a six-week production horizon.
- Built four trigger pipelines on the shared schema (cron briefing, threshold material forecast, calendar webhook, and a human approval gate), with a rejection branch that returns invalid schedule changes with a reason instead of corrupting state.
- Kept scheduling decisions deterministic and constraint-checked for auditability, using an LLM only to narrate finished decisions into plain-language briefings.

Inbox Orchestration | *n8n, Outlook, LM Studio, Qwen2.5-7B, Excel, Discord*

2026

- Built a multi-agent pipeline that routes every incoming email through a local 7B orchestrator, which classifies each message and drops marketing and noise before invoking a specialist agent.
- Designed an approval gate so drafted replies post to Discord and wait for a human decision before anything is sent, and a second agent that extracts company, role, status, and date from job-related email and appends structured rows to a tracking sheet.
- Ran the system against 1,000 real messages, logging 851 appended rows and three concrete failure modes (duplicate rows, hallucinated fields, unnecessary tool calls) to scope the next fix.

Multi-Agent UAV Path Planning | *PyTorch, MAPPO, NumPy, NVIDIA Isaac Sim*

2026

- Trained an 8-agent coordination policy using MAPPO with a centralized critic and a parameter-shared Gaussian actor, applying centralized training with decentralized execution so each agent acts on local observations alone.
- Combined global A* routing with reciprocal velocity obstacle avoidance, and built an independent collision auditor that scores every frame without steering, making violations measurable failures rather than tuning artifacts.

EXPERIENCE

Packer

Multi-Plastics Extrusion

Hazleton, PA

May 2025 – Sep. 2025

- Packed and staged finished film rolls on 12-hour night shifts, inspecting output for defects and flagging quality issues under continuous throughput targets.

Resident Assistant

Penn State University

Hazleton, PA

Aug. 2023 – May 2024

- Supported 40+ first-year residents through conflict resolution, mentorship, and community events; coordinated with housing staff on policy enforcement and campus resource referrals.